

SECTION 8 – ELECTRICAL CONNECTIONS FOR RAMPING AND SHIMMING

Description:

Main and Superconducting Shim Power Supply Input / Output Connections have the same (P/J) designations, used in this procedure, for both the Service and Phase III Power Supply Units.

Make sure that the Power Supply Cabinets are installed and checked out in conformance with the supplier manual, supplied with the unit, before making any power supply connections to the magnet. See INTRODUCTION, Section 2 (“Vendor Manuals”) for supplier manual numbers.

SCHEMATICS/INTERCONNECTS, Section 1–1 (“Magnet Interconnect Diagram”) shows the Cable Numbers, Run Numbers and Interconnect Pattern covered in this section.

If Ramp and Shim Cables were not ordered with the magnet, Field Ramp Cable Kit P/N 2135435 will be needed to ramp the magnet and Field Shim Cable Kit P/N 2135558 will be needed to shim the magnet. If Ramp and Shim Cables will remain on site, order the appropriate R–Cat numbers as explained in Service Note 63034.

Procedure:

8–1 SUPERCONDUCTING SHIM COIL POWER SUPPLY CONNECTIONS



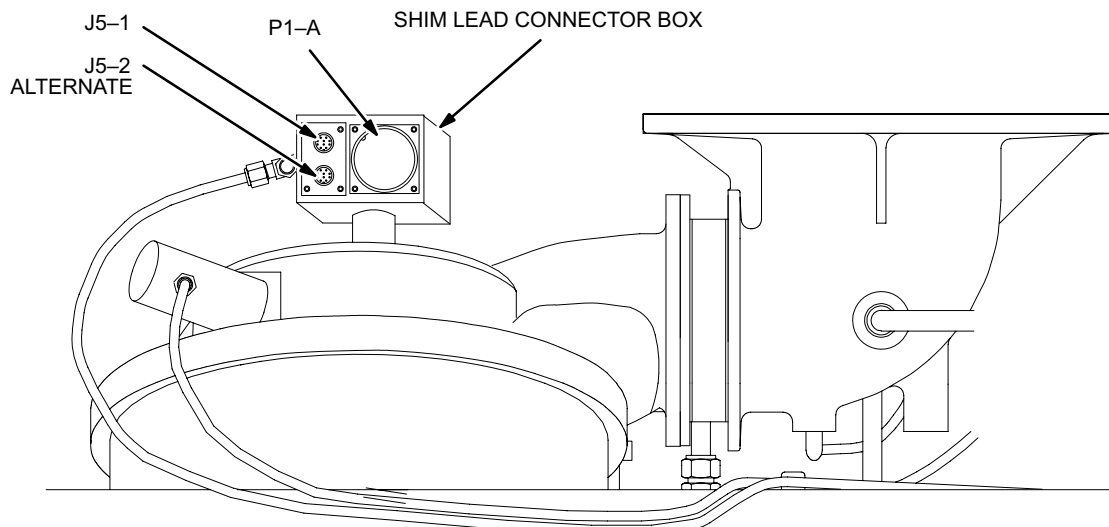
DO NOT CONNECT INPUT POWER TO OR TURN ON SUPERCONDUCTING SHIM COIL POWER SUPPLY UNTIL IT IS VERIFIED THAT ALL CURRENT CONTROLS ARE SET AT ZERO, FULLY COUNTERCLOCKWISE.

CONNECTING THE SUPERCONDUCTING SHIM COIL HARNESS TO THE MAGNET WHEN THE SUPERCONDUCTING SHIM COIL POWER SUPPLY IS ON MAY CAUSE IRREPARABLE DAMAGE TO THE VAPOR COOLED SHIM LEADS.

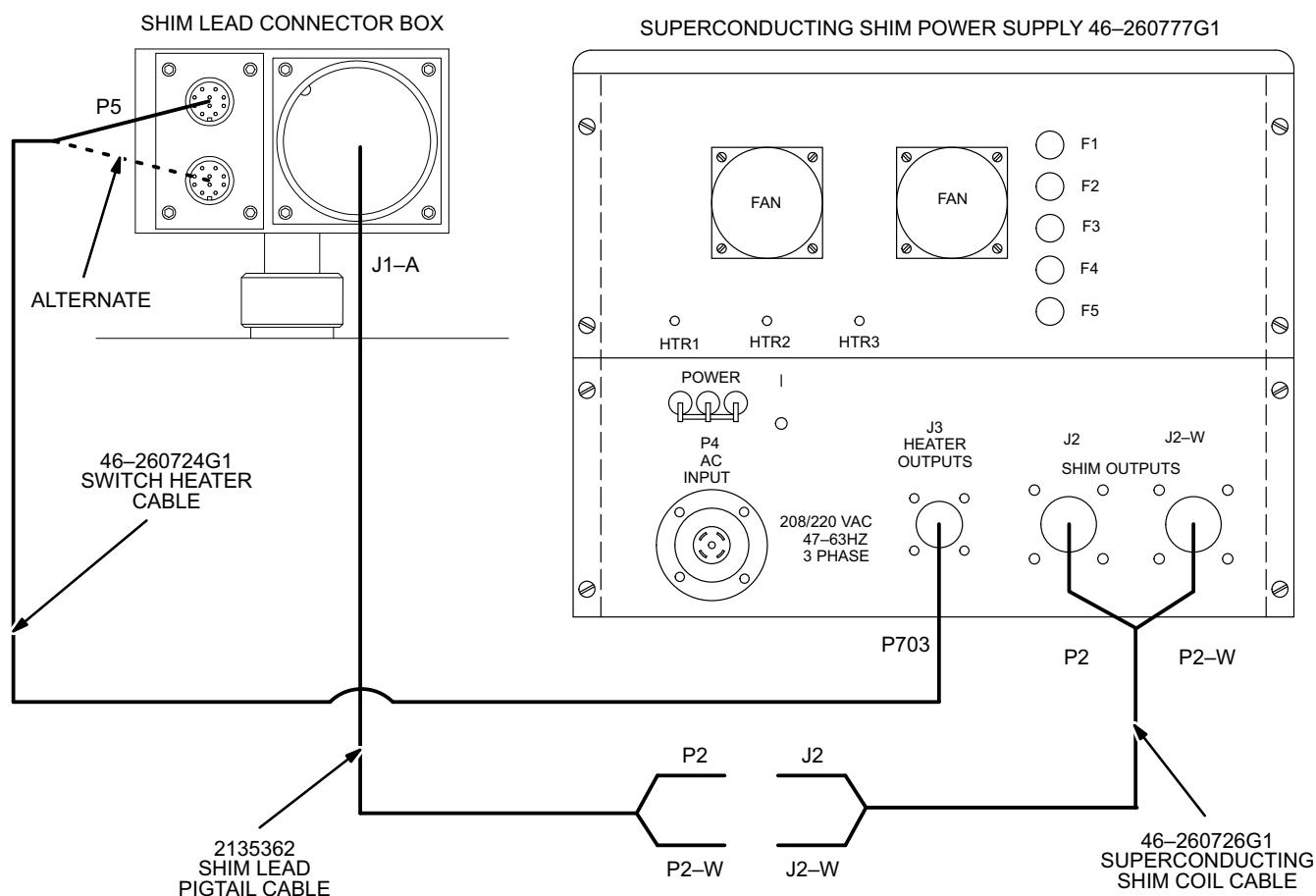


Never place weight (objects) on the coldhead sleeve as irreparable damage may result.

1. Verify that input power to the Superconducting Shim Coil Power Supply is disconnected.
2. Connect electrical cabling to the Superconducting Shim Coil Power Supply as shown in Illustrations 8–1 and 8–2.

8-1 SUPERCONDUCTING SHIM COIL POWER SUPPLY CONNECTIONS (continued)

MAGNET ELECTRICAL CONNECTIONS
ILLUSTRATION 8-1



SHIM POWER SUPPLY OUTPUT CONNECTIONS
ILLUSTRATION 8-2

8-2 MAIN COIL POWER SUPPLY CONNECTIONS

1. Verify that the input power to the Main Coil Power Supply is disconnected.
2. Connect electrical cabling to the Main Coil Power Supply as shown in Illustrations 8-1, 8-3 and 8-4.

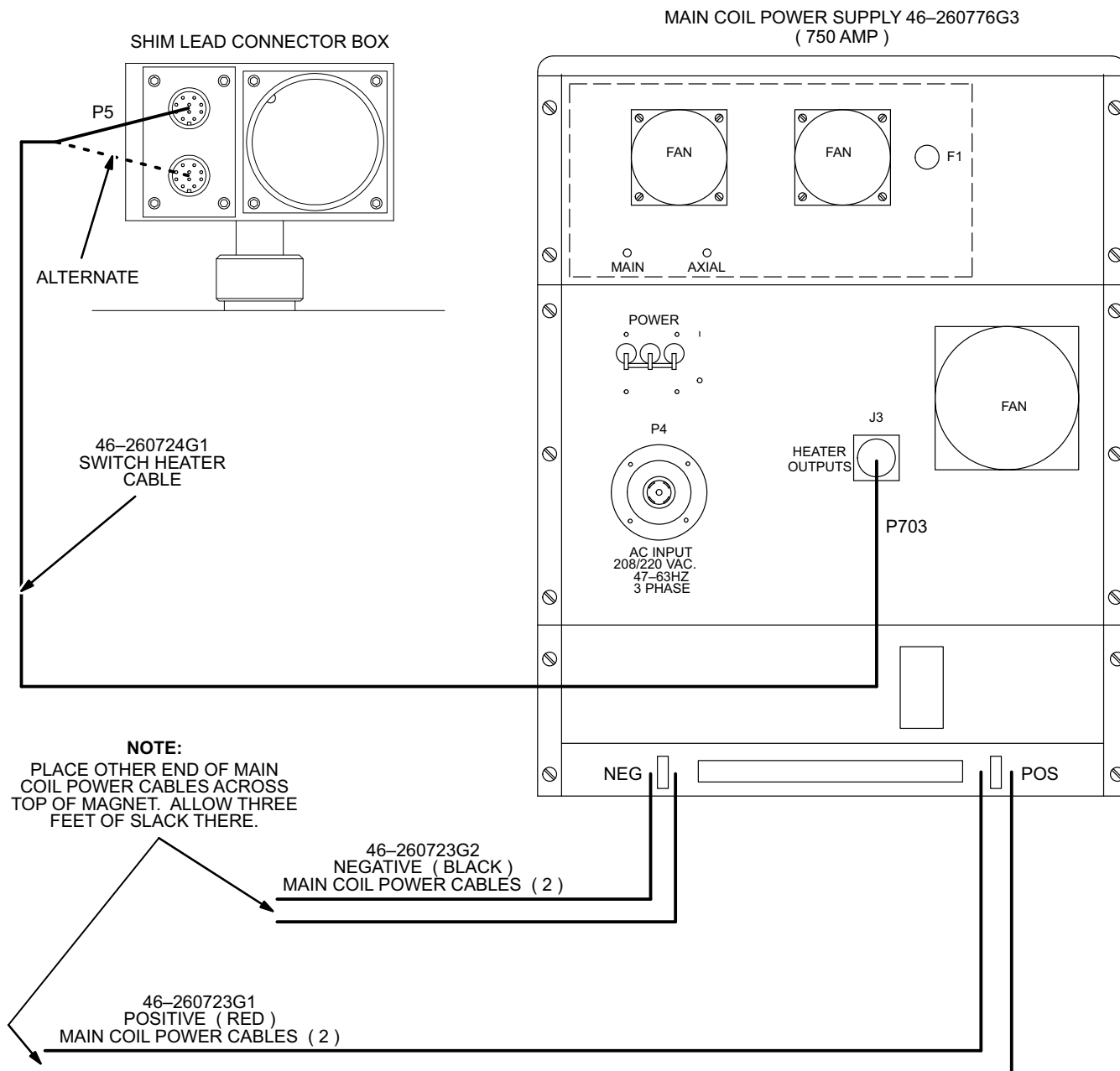
Note

Clean all connection points on Power Leads and Power Lead Extensions to prevent high resistance contacts and minimize voltage drops. A scouring pad can be used to clean all connections.

Make sure the nuts are tightened sufficiently to prevent a high resistance contact. Connect Red Cables to the positive (+) Buss Bar and Black Cables to the negative (-) Buss Bar. Two Red and two Black Cables are connected in parallel to the Buss Bars. Make sure that the Cable Lugs and / or exposed wire from the Ramp Cables are not touching the case of the Main Coil Power Supply.

3. Check Cryostat Pressure Gauge. If the pressure is above 0.25 psi, slowly open Helium Vent Valve (V2) and allow pressure to drop to 0.25 psi. Then close valve (V2).

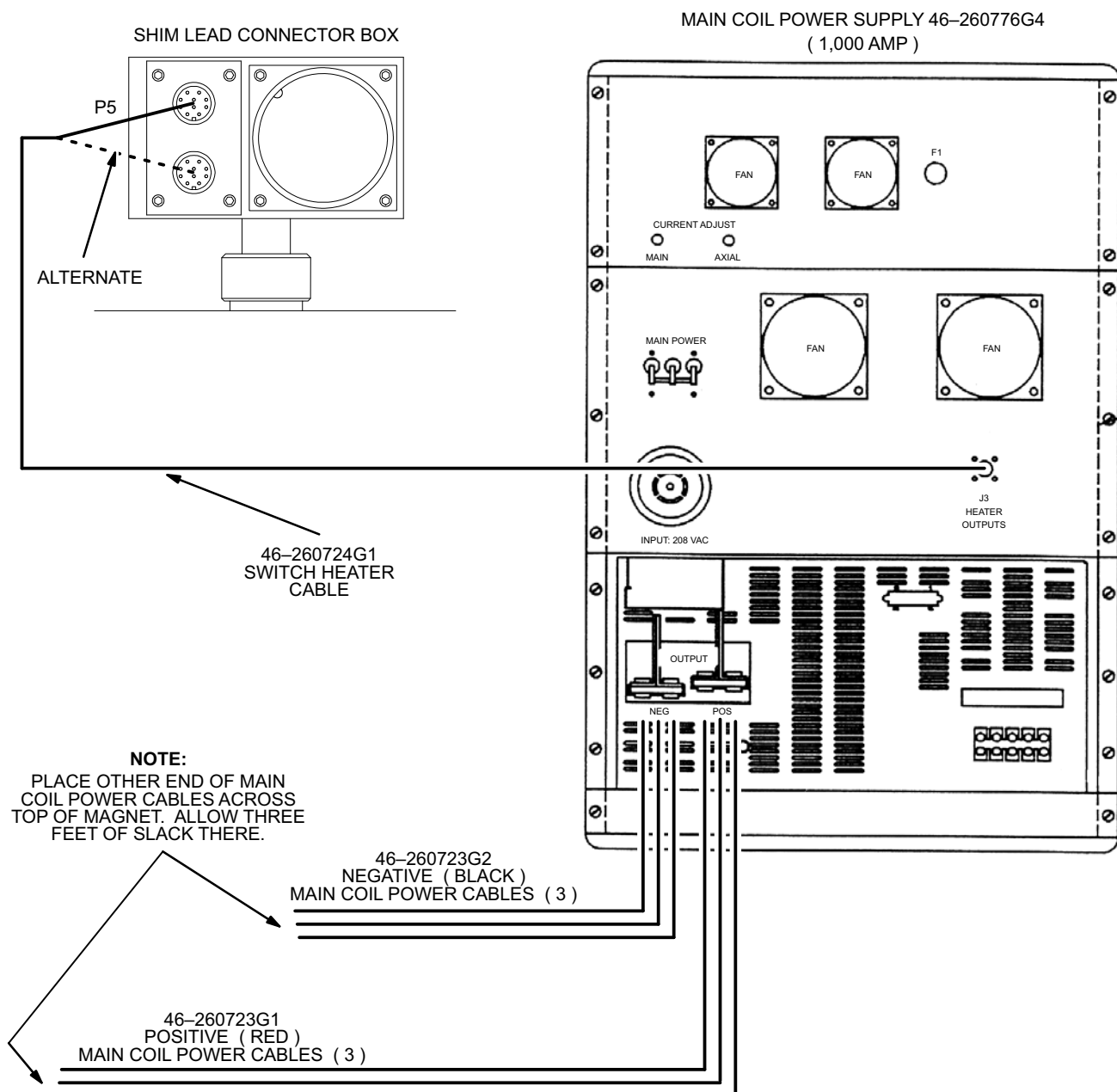
8-2 MAIN COIL POWER SUPPLY CONNECTIONS (continued)



MAGNET POWER SUPPLY CONNECTIONS (750 AMP)

ILLUSTRATION 8-3

8-2 MAIN COIL POWER SUPPLY CONNECTIONS (continued)



MAGNET POWER SUPPLY CONNECTIONS (1,000 AMP)

ILLUSTRATION 8-4

8-2 MAIN COIL POWER SUPPLY CONNECTIONS (continued)

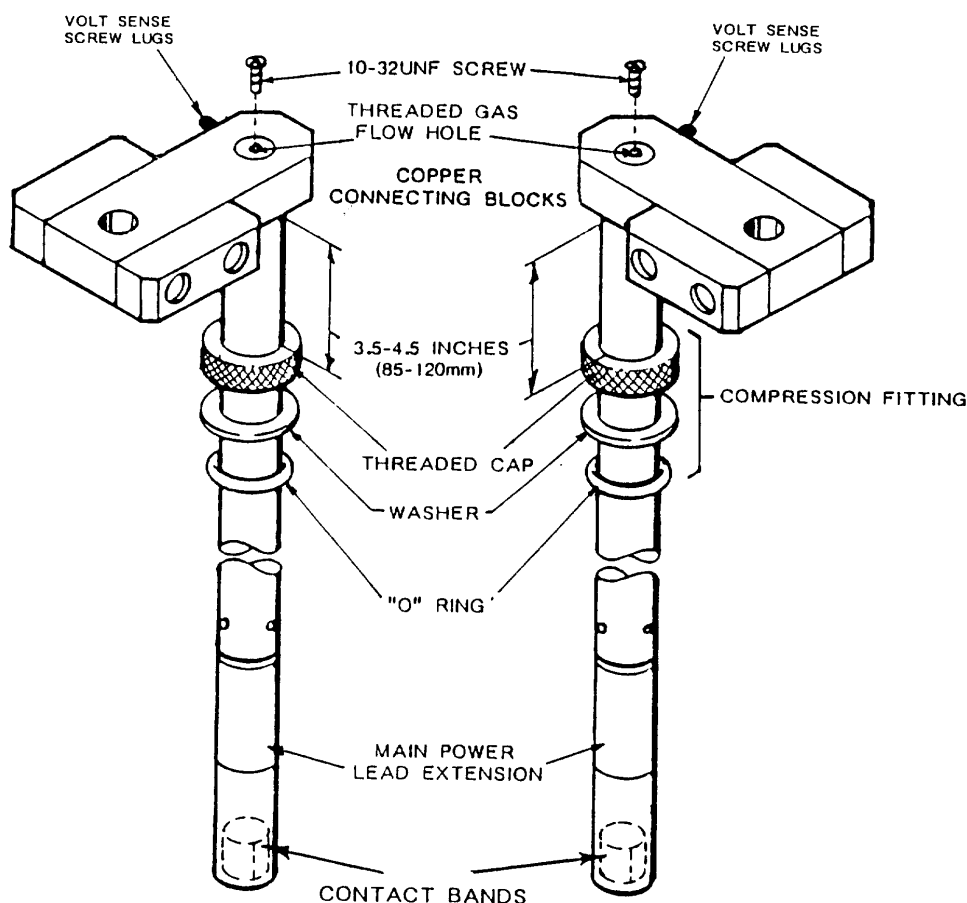


The Main Power Lead Extensions for the magnet are different from other magnets. The Lead Extensions can be recognized by a Helium Flow Hole as viewed from the Contact Band End of each Extension. The Ramp Lead Extensions are also shorter than the SIII Ramp Leads. Make sure the Power Lead Extensions (46-294204G1) are used for this procedure.

Note

Use a new set of Contact Bands for each ramp performed.

4. Replace Contact Bands on Main Power Lead Extensions. Make sure gas flow holes are not blocked. See REPLACEMENT / MAINTENANCE, Section 12.
5. Remove the Threaded Caps, Washers and "O" Rings (46-294104P1, 46-294105P1 & 46-260389P1) from the plastic bag, taped to the Shim Lead Assembly in a plastic bag, and mount them on the Main Power Lead Extensions. See Illustration 8-5.



COMPRESSION FITTING MOUNTING ON MAIN POWER LEAD EXTENSION

ILLUSTRATION 8-5

8-2 MAIN COIL POWER SUPPLY CONNECTIONS (continued)**WARNING!**

MAKE SURE MAGNET ROOM VENT EXHAUST FAN IS TURNED ON, OR THE HATCH IS OPENED IF A MOBILE VAN, BEFORE STARTING THIS PROCEDURE. THIS IS REQUIRED TO EXHAUST THE ODORLESS AND INVISIBLE HELIUM GAS GENERATED DURING THIS PROCEDURE AND PREVENT OXYGEN DISPLACEMENT IN THE MAGNET ROOM. REVIEW AND FOLLOW CRYOGEN SAFETY MEASURES CONTAINED IN SECTION 5-3 OF THE INTRODUCTION (CRYOGEN SAFETY).

IF THE MAGNET IS RAMPED UP, WAIT FOR THE LEAD EXTENSIONS TO COOL SUFFICIENTLY (A FOG OR WATER VAPOR FORMS AROUND THE LEAD EXTENSIONS) BEFORE FULLY ENGAGING THEM TO PREVENT THE RISK OF QUENCHING THE MAGNET.

A SUPERCONDUCTING MAGNET AT FIELD IS A HIGH ENERGY STORAGE DEVICE CAPABLE OF DISCHARGING RAPIDLY. DO NOT TOUCH THE MAIN LEAD EXTENSIONS SIMULTANEOUSLY OR ALLOW THEM TO COME IN CONTACT WITH EACH OTHER WHEN THE MAGNET IS BEING RAMPED OR AT FIELD AS A RAPID DISCHARGE WOULD RESULT THROUGH THEIR CONTACT POINTS IF THE SWITCH HEATER BECOMES ACTIVATED OR CIRCUIT RESISTANCE DEVELOPS.

WARNING!

TO PREVENT THE POSSIBILITY OF MIS-WIRING AND A RESULTANT QUENCH DURING FUTURE RAMPING OF THE MAGNET, CONNECTION POLARITIES MUST BE NOTED AND RECORDED IN THE DATA SHEET TAB, TABLE 6-1.

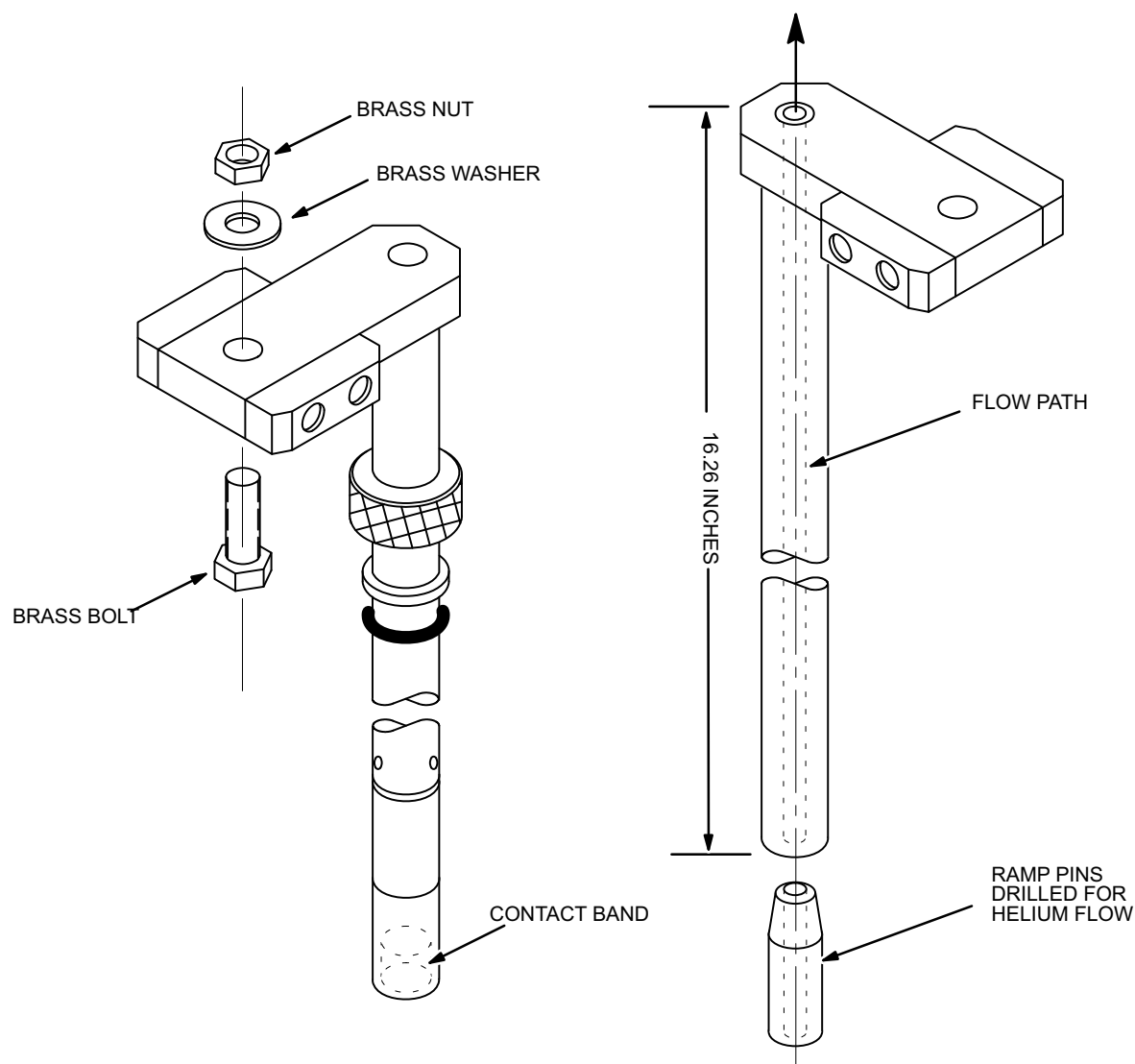
Note

Use only Brass Washers when connecting Main Power Leads to the Main Lead Extensions to keep voltage drops to a minimum.

CAUTION

Make sure the Brass Bolts in Step 6 are installed as shown in Illustration 8-6 to prevent any possibility of contact with the Fill Port Cap during ramping which would quench the magnet.

6. Connect the other end of the Main Power Cables to the main Power Lead Extensions with the 1 inch (25.4mm) Brass Bolts as shown in Illustration 8-6. Secure bolts with the Brass Nuts and Brass Washers provided. Tighten the connections sufficiently to provide a good electrical connection.
7. Install the 10-32UNF Gas Flow Hole Screws into the gas flow holes on each Ramp Lead Extension. See Illustration 8-5.

8-2 MAIN COIL POWER SUPPLY CONNECTIONS (continued)

HARDWARE MOUNTING ON RAMP LEAD EXTENSIONS
ILLUSTRATION 8-6

8-2 MAIN COIL POWER SUPPLY CONNECTIONS (continued)

The "Normal" ramp polarity is "+" on the left and "-" on the right as viewed from the cold head side of the magnet. "Reversed" ramp polarity has "+" on the right and "-" on the left as viewed from the cold head side of the magnet. If using the Polarity Plate, make sure that the Polarity Plate properly indicates which polarity the magnet is to be ramped. Make sure the chosen ramp polarity is recorded in the DATA SHEET tab, Section NO TAG.

Note

The Ramp Polarity Plate position is changed by first venting the magnet to below 0.5 psi, then removing both ramp port caps. Pull the polarity plate up over the ramp ports and reinstall the plate to indicate the chosen polarity. Replace Ramp Port Caps.

8. The Optional Polarity Plate, P/N 2111737, is normally used only to indicate a reverse ramped magnet. The plate can be ordered if desired. To install, open Vent Valve V2 until the magnet pressure is around 0.2 psi. Then quickly unscrew both Ramp Port Caps and place the Polarity Plate, with the desired Ramp Polarity indicated, over the Ramp Port openings. See Illustration 8-7 for Reverse Ramp Indication.

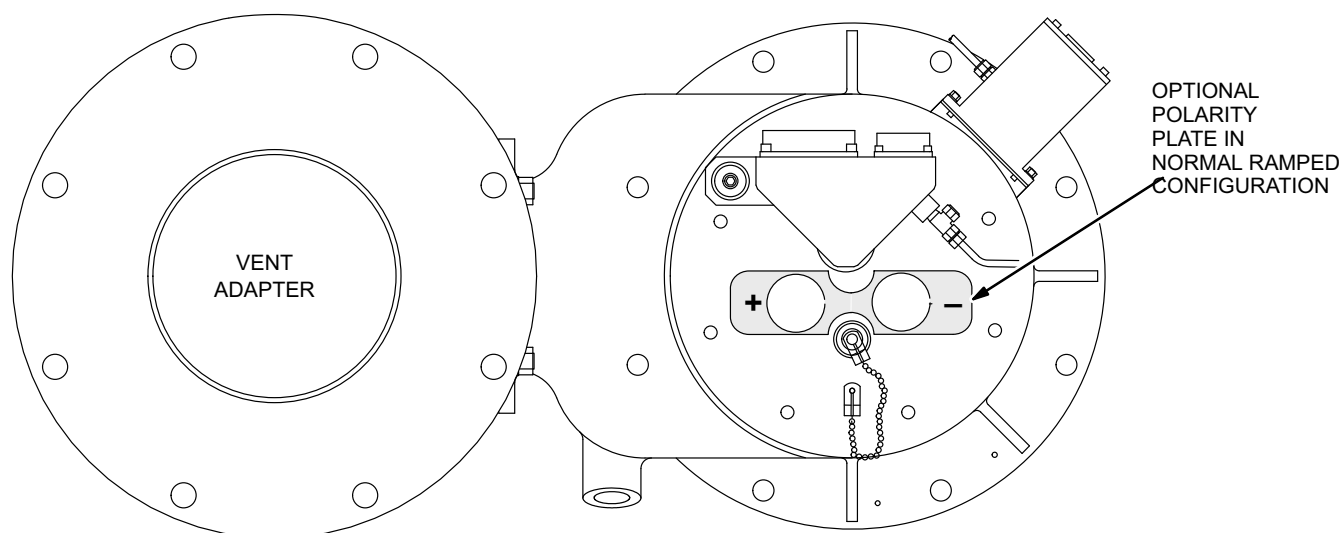
**RAMP LEAD POLARITY**

ILLUSTRATION 8-7

9. Remove the cap of the "+" Main Power Lead Extension Receptacles located on the vertical stack of the magnet. Make sure that the gasket inside the cap does not get lost. See Illustration 8-7.

Note

Screw Main Lead Extension Compression Nut onto the Ramp Port, quickly, to prevent icing of threads.

8-2 MAIN COIL POWER SUPPLY CONNECTIONS (continued)**Note**

For Reverse Ramped magnet, reverse the connections in Step 10 (i.e., insert the Positive (red cables attached) Ramp lead Extension into the “-” Ramp Lead Port and insert the Negative (black cables attached) into the “+” Ramp Lead Port).

10. Quickly insert the Positive (red cables attached) Main Power Lead Extension, about halfway, into the open receptacle. Loosely screw the Compression Nut onto the Ramp Lead Port. See Illustration 8-7 for correct connection polarities.
11. Repeat Steps 9 through 10 for the other Lead Extension.
12. Connect the volt sense leads to the Main Power Lead Extensions. See Illustration 8-8. Terminate other end of volt sense leads to a DVM or VOM placed near the Main Coil Power Supply.
13. Remove Flow Hole Screws. See Illustration 8-8.

Note

Lead Extensions will depress approximately 1 inch (25mm) from the point of contact to the fully engaged position. A firm contact will be felt when fully engaged. Do not rotate Lead Extensions excessively when in the engaged position as internal contact wear could result.

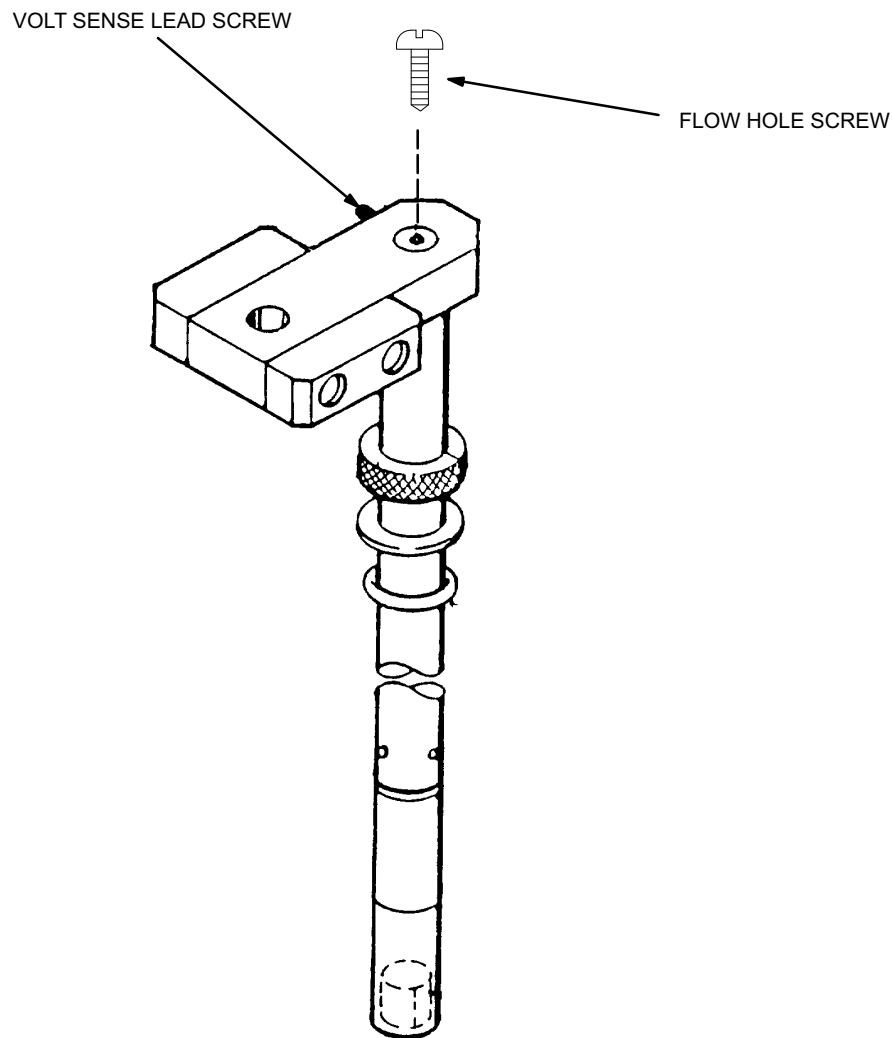


IF THE MAGNET IS RAMPED UP, WAIT FOR THE LEAD EXTENSIONS TO COOL SUFFICIENTLY (A FOG OR WATER VAPOR FORMS AROUND THE LEAD EXTENSIONS) BEFORE FULLY ENGAGING THEM, TO PREVENT THE RISK OF QUENCHING THE MAGNET.

14. When the Lead Extensions are sufficiently cooled, loosen the Compression Fittings and fully engage the Lead Extensions. Loosely screw the Compression Nut onto the Ramp Lead Ports.



Make sure gas flow holes in Lead Extension are not blocked and GHe is exiting holes.

8-2 MAIN COIL POWER SUPPLY CONNECTIONS (continued)

VOLT SENSE LEAD CONNECTIONS
ILLUSTRATION 8-8



During ramping, high currents in the power cables may cause movement of the cables due to magnetic forces. Excess motion may disrupt contact between the ramp probes and their contact pins causing a quench. Take the following steps to prevent lead motion.

15. After installing the ramp leads, secure the main power cables using Ty Wraps (field supplied item). The cables should be secured to each other and to convenient fixed points, such as the shroud rails or the magnet lifting rings, to prevent their movement.

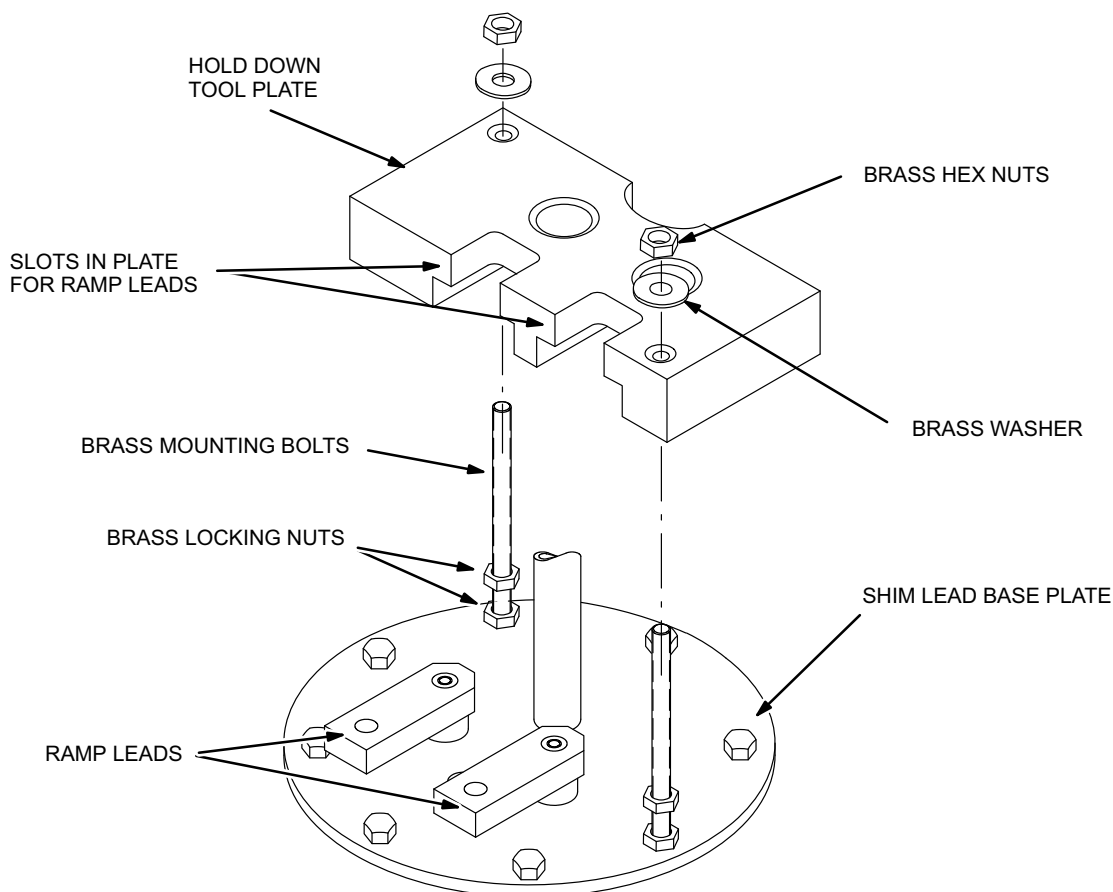
8-2 MAIN COIL POWER SUPPLY CONNECTIONS (continued)

16. A Ramp Lead Hold Down Tool (2185743), is a mandatory tool which will minimize lead motion and aid in getting good contact resistance. Install the tool as follows: Remove two bolts from the Shim Lead Base Plate on top of the magnet (Save these bolts to be put back later). Screw the Mounting Bolts of the Hold Down Tool into these mounting holes. See Illustration 8-9.
17. Install the Hold Down Tool Plate onto the mounting bolts. Make sure that the ramp leads fit into the slots on the plate. See Illustration 8-9.



Make sure that the vent holes in the ramp leads are not blocked. This is necessary to minimize the Ramp Lead to Ramp Pin voltage drop, thereby, eliminating a quench potential.

18. Tighten the Nuts on top of the Hold Down Tool Plate to lock the plate firmly in place and prevent lead motion. While tightening the plate nuts, adjust the leveling screw to keep the plate level. See Illustration 8-9.



HOLD DOWN TOOL INSTALLATION
ILLUSTRATION 8-9